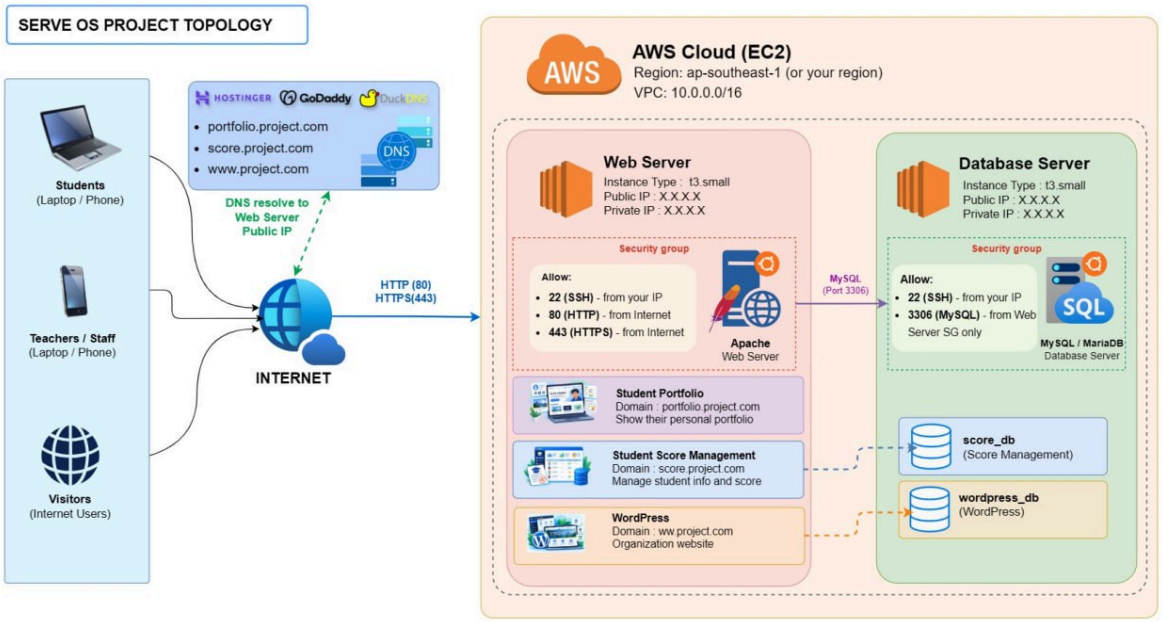
PROJECT SERVER OS

Build, Deploy, Secure & Monitor a Web Server

Topology



Create a security group

- Web-server

Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
SSH	TCP	22	My IP	123.108.252.178/32		Delete
HTTP	TCP	80	Anywh...	0.0.0.0/0		Delete
HTTPS	TCP	443	Anywh...	0.0.0.0/0		Delete

Add rule

- Database-server

Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
SSH	TCP	22	My IP	123.108.252.178/32		Delete
MYSQL/Aurora	TCP	3306	Anywh...	0.0.0.0/0		Delete

Add rule

Hosting for My Portfolio

1. Install apache2 service

```
ubuntu@webserver:~$ sudo apt install apache2 -y
Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64  liblua5.4-0  ssl-cert

Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom  www-browser

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 186
  Download size: 2123 kB
  Space needed: 8209 kB / 17.2 GB available
```

2. Git clone

```
ubuntu@webserver:~$ sudo git clone https://github.com/riengnarong-alt/my-portfolio.git                    https://github.com/riengnarong-alt/my-portfolio.git
Cloning into 'my-portfolio'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), 262.60 KiB | 26.26 MiB/s, done.
```

```
git clone https://github.com/riengnarong-alt/my-portfolio.git
```

3. Create directory

```
ubuntu@webserver:~$ sudo mkdir -p /var/www/portfolio
ubuntu@webserver:~$ ls portfolio
ls: cannot access 'portfolio': No such file or directory
ubuntu@webserver:~$ ls -la portfolio
ls: cannot access 'portfolio': No such file or directory
ubuntu@webserver:~$ ls -ld /var/www/portfolio
drwxr-xr-x 2 root root 4096 Oct  5 04:14 /var/www/portfolio
ubuntu@webserver:~$ cd /var/www/
ubuntu@webserver:/var/www$ cd portfolio
ubuntu@webserver:/var/www/portfolio$
```

```
sudo mkdir -p /var/www/portfolio
ls -ld /var/www/portfolio
```

- **Copy file**

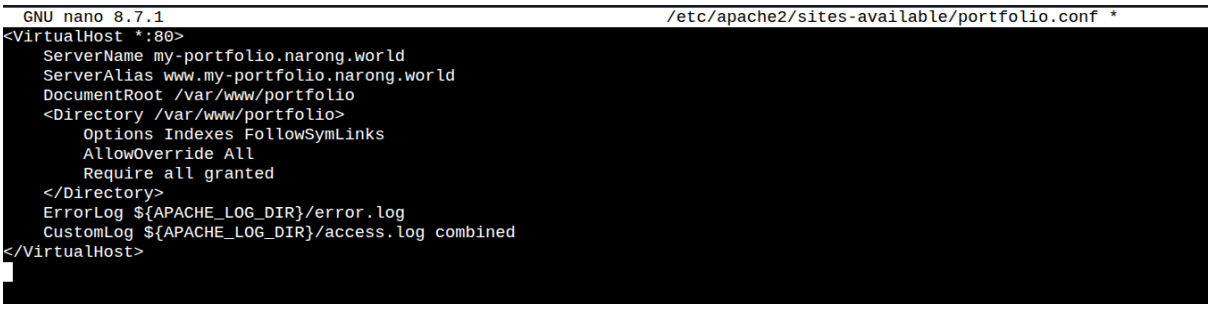
```
ubuntu@webserver:/var/www/portfolio$ sudo cp -r ~/my-portfolio/* /var/www/portfolio/
ubuntu@webserver:/var/www/portfolio$ sudo chown -R www-data:www-data /var/www/portfolio/
ubuntu@webserver:/var/www/portfolio$ ls -la /var/www/portfolio

total 288
drwxr-xr-x 2 www-data www-data   4096 Oct  5 04:22  .
drwxr-xr-x 4 root     root       4096 Oct  5 04:14  ..
-rw-r--r-- 1 www-data www-data 261418 Oct  5 04:22  'MY PIC.png'
-rw-r--r-- 1 www-data www-data  20802 Oct  5 04:22  index.html
ubuntu@webserver:/var/www/portfolio$
ubuntu@webserver:/var/www/portfolio$
```

```
sudo cp -r ~/my-portfolio/* /var/www/portfolio/
sudo chown -R www-data:www-data /var/www/portfolio/
ls -la /var/www/portfolio
```

4. Create virtual hosts

```
sudo nano /etc/apache2/sites-available/portfolio.conf
```




```
<VirtualHost *:80>
    ServerName my-portfolio.narong.world
    ServerAlias www.my-portfolio.narong.world
    DocumentRoot /var/www/portfolio
    <Directory /var/www/portfolio>
        Options -Indexes +FollowSymLinks
        AllowOverride None
        Require all granted
    </Directory>
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

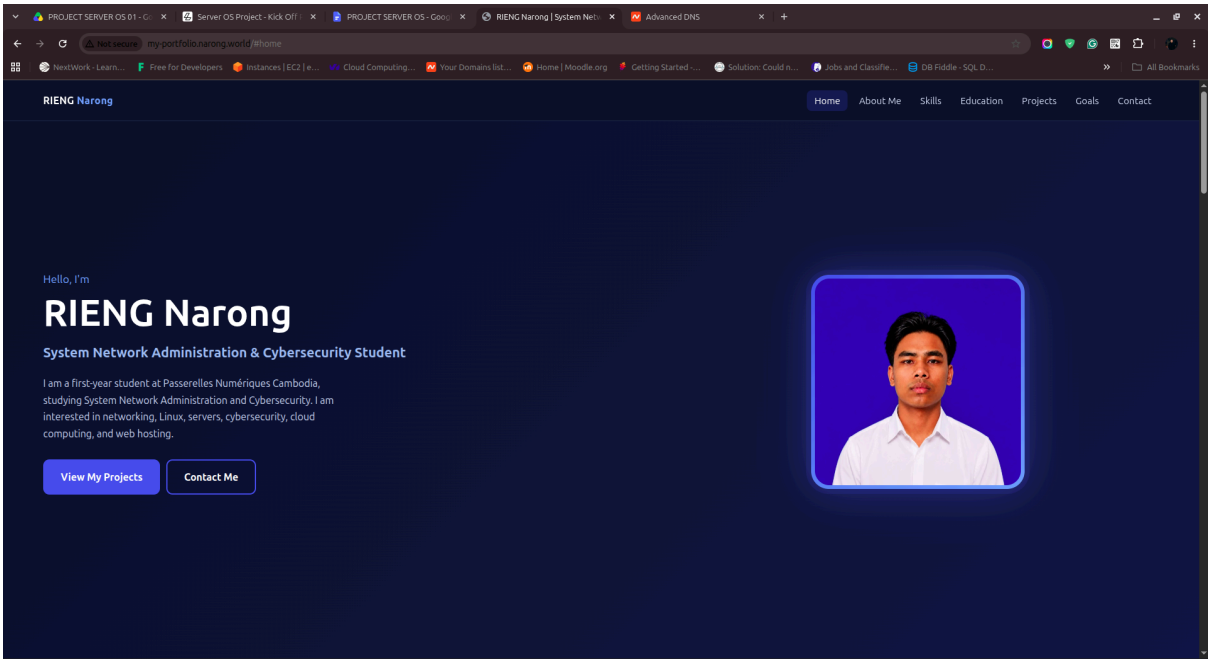


- **Checklist before this goes live:**

```
cd /etc/apache2/sites-available
sudo a2dissite 000-default.conf
sudo a2ensite portfolio.conf
sudo systemctl reload apache2
```

- **Then verify:**

```
curl -H "Host: my-portfolio.narong.world" http://localhost
```



http://my-portfolio.narong.world

+ Change my-portfolio to HTTPS

```
ubuntu@webserver:/etc/apache2/sites-available$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-05 03:22:09 UTC; 1h 51min ago
 Invocation: 0c781d7165654984b56e40f335f97e41
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 4618 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 2968 (apache2)
     Status: "Total requests: 69; Idle/Busy workers 100/0;Requests/sec: 0.0103; Bytes served/sec:  52 B/sec"
      Tasks: 55 (limit: 1741)
     Memory: 6.1M (peak: 9.7M)
        CPU: 600ms
     CGroup: /system.slice/apache2.service
             ├─2968 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─4622 /usr/sbin/apache2 -k start -DFOREGROUND
             └─4623 /usr/sbin/apache2 -k start -DFOREGROUND

Oct 05 03:22:09 webserver systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 05 03:22:09 webserver systemd[1]: Started apache2.service - The Apache HTTP Server.
Oct 05 05:04:46 webserver systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Oct 05 05:04:46 webserver systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
ubuntu@webserver:/etc/apache2/sites-available$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
OpenSSH                    ALLOW       Anywhere
Apache Full                ALLOW       Anywhere
OpenSSH (v6)               ALLOW       Anywhere (v6)
Apache Full (v6)           ALLOW       Anywhere (v6)

ubuntu@webserver:/etc/apache2/sites-available$
ubuntu@webserver:/etc/apache2/sites-available$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
```

```
sudo systemctl status apache2
sudo ufw status

sudo apt update
sudo apt install -y certbot python3-certbot-apache
```

```
ubuntu@webserver:/etc/apache2/sites-available$ sudo certbot --apache -d my-portfolio.narong.world
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for my-portfolio.narong.world

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/my-portfolio.narong.world/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/my-portfolio.narong.world/privkey.pem
This certificate expires on 2027-01-03.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for my-portfolio.narong.world to /etc/apache2/sites-available/portfolio-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://my-portfolio.narong.world

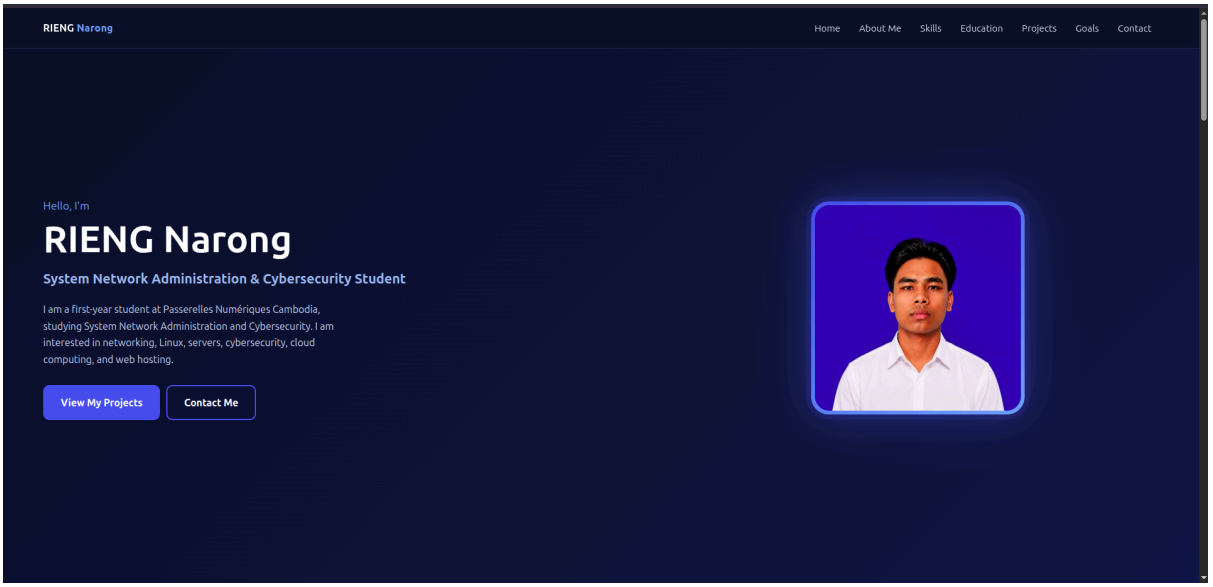
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
```

sudo certbot --apache -d my-portfolio.narong.world -d www.my-portfolio.narong.world

sudo certbot --apache -d my-portfolio.narong.world

```
ubuntu@webserver:/etc/apache2/sites-available$ curl -I https://my-portfolio.narong.world
HTTP/1.1 200 OK
Date: Mon, 05 Oct 2026 05:26:57 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Mon, 05 Oct 2026 04:22:33 GMT
ETag: "5142-65d103b29f272"
Accept-Ranges: bytes
Content-Length: 20802
Vary: Accept-Encoding
Content-Type: text/html
```

curl -I https://my-portfolio.narong.world



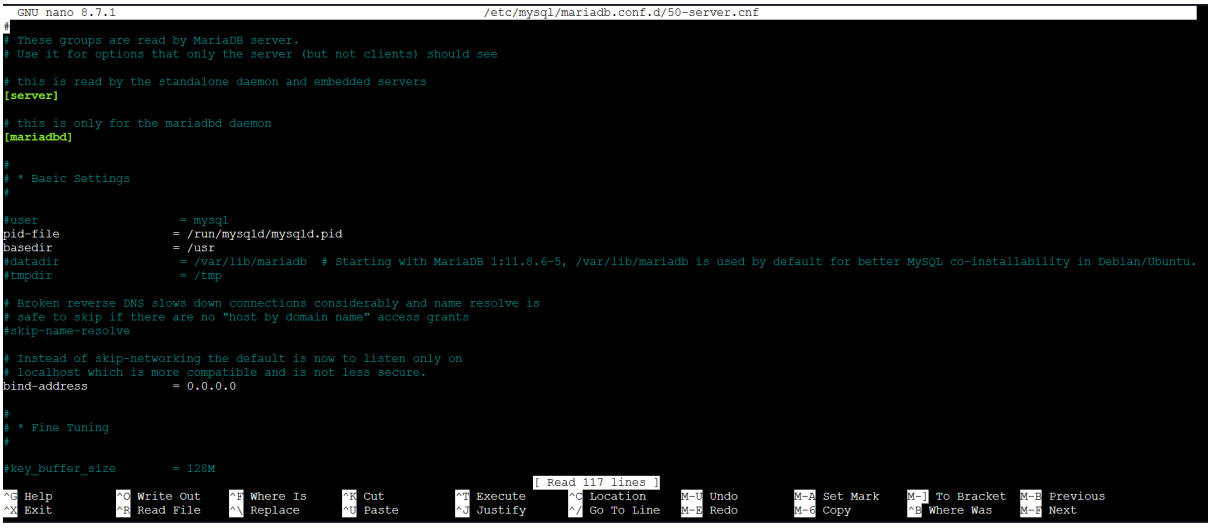
https://my-portfolio.narong.world/#home

Done for my portfolio

Hosting for Student Score Management

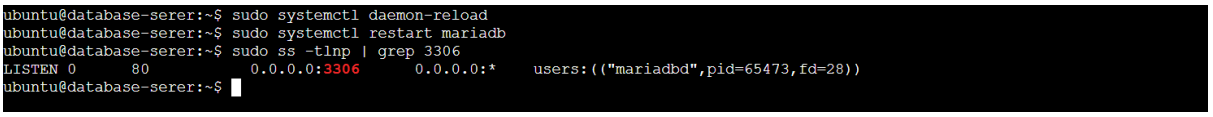
(Database-server)

1. Allow network connections




```
sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf
# Set:  bind-address = <Database-server private IP>   (do not use 0.0.0.0)
# Then: sudo systemctl restart mariadb
```

+ Verify it's listening on the network




```
sudo ss -tlnp | grep 3306
```

2. Create a database + user for the web server

```
ubuntu@database-serer:~$ sudo mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.8.6-MariaDB-5ubuntu0.1 from Ubuntu -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> DROP USER IF EXISTS 'score_user'@'%';
Query OK, 0 rows affected, 1 warning (0.003 sec)

MariaDB [(none)]> DROP DATABASE IF EXISTS score_db;
Query OK, 0 rows affected, 1 warning (0.000 sec)

MariaDB [(none)]> CREATE DATABASE score_db;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'score_user'@'172.31.29.28' IDENTIFIED BY 'strong';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON score_db.* TO 'score_user'@'172.31.29.28';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)
```

```
DROP USER IF EXISTS 'score_user'@'%';
DROP DATABASE IF EXISTS score_db;
CREATE DATABASE score_db;
CREATE USER 'score_user'@'172.31.29.28' IDENTIFIED BY 'strong';
GRANT SELECT, INSERT, UPDATE, DELETE, CREATE, ALTER, DROP, INDEX,
REFERENCES ON score_db.* TO 'score_user'@'172.31.29.28';
FLUSH PRIVILEGES;
```

3. Open the firewall for port 3306

```
ubuntu@database-serer:~$ sudo ufw allow from 172.31.29.28 to any port 3306
Rules updated
ubuntu@database-serer:~$ sudo ufw status
Status: inactive
ubuntu@database-serer:~$
```

```
sudo ufw allow from 172.31.29.28 to any port 3306
sudo ufw status
```

(Web-server)

1. Install prerequisites

```
ubuntu@webserver:~$ sudo apt install composer php-cli php-mbstring php-xml php-curl php-zip php-mysql php-sqlite3 php-fpm nodejs npm
php-cli is already the newest version (2:8.5+99ubuntu1).
php-mbstring is already the newest version (2:8.5+99ubuntu1).
php-xml is already the newest version (2:8.5+99ubuntu1).
php-curl is already the newest version (2:8.5+99ubuntu1).
php-zip is already the newest version (2:8.5+99ubuntu1).
php-mysql is already the newest version (2:8.5+99ubuntu1).
php-fpm is already the newest version (2:8.5+99ubuntu1).
nodejs is already the newest version (22.23.3-1nodesource1).
Solving dependencies... Error!
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
The following information may help to resolve the situation:

Unsatisfied dependencies:
 composer : Depends: php-composer-ca-bundle (>= 1.5) but it is not going to be installed
            Depends: php-composer-ca-bundle (< 2~~) but it is not going to be installed
            Depends: php-composer-class-map-generator (>= 1.4.0) but it is not going to be installed
            Depends: php-composer-class-map-generator (< 2~~) but it is not going to be installed
            Depends: php-composer-metadata-minifier (>= 1.0) but it is not going to be installed
            Depends: php-composer-metadata-minifier (< 2~~) but it is not going to be installed
            Depends: php-composer-semver (>= 3.3) but it is not going to be installed
            Depends: php-composer-semver (< 4~~) but it is not going to be installed
            Depends: php-composer-spdx-licenses (>= 1.5.7) but it is not going to be installed
            Depends: php-composer-spdx-licenses (< 2~~) but it is not going to be installed
            Depends: php-composer-xdebug-handler but it is not going to be installed
            Depends: php-json-schema (>= 6.5.1) but it is not going to be installed
            Depends: php-json-schema (< 7~~) but it is not going to be installed
            Depends: php-psr-log but it is not going to be installed
            Depends: jsonlint (>= 1.4) but it is not going to be installed
```

```
sudo apt update
sudo apt install composer php-cli php-mbstring php-xml php-curl
php-zip php-mysql php-sqlite3 php-fpm nodejs npm
```

+ Verify version

```
ubuntu@webserver:~$ composer --version
sudo npm install -g npm@latest
npm -v
node -v
Composer version 2.10.3 2026-08-27 13:34:23
PHP version 8.5.4 (/usr/bin/php8.5)
Run the "diagnose" command to get more detailed diagnostics output.

removed 65 packages, and changed 91 packages in 5s

15 packages are looking for funding
  run `npm fund` for details
12.2.0
v22.23.3
ubuntu@webserver:~$
```

```
composer --version
sudo npm install -g npm@latest
npm -v
node -v
```

2. Clone repo (Backend)

```
ubuntu@webserver:~$ cd /var/www
ubuntu@webserver:/var/www$ sudo git clone https://github.com/riengnarong-alt/Student-Score-Management-Backend.git
Cloning into 'Student-Score-Management-Backend'...
remote: Enumerating objects: 245, done.
remote: Counting objects: 100% (245/245), done.
remote: Compressing objects: 100% (175/175), done.
remote: Total 245 (delta 53), reused 242 (delta 53), pack-reused 0 (from 0)
Receiving objects: 100% (245/245), 325.04 KiB | 5.42 MiB/s, done.
Resolving deltas: 100% (53/53), done.
```

```
cd /var/www
sudo git clone
https://github.com/riengnarong-alt/Student-Score-Management-Backend.git
```

+ Nested folder

```
ubuntu@webserver:/var/www$ cd /var/www/Student-Score-Management-Backend
ubuntu@webserver:/var/www/Student-Score-Management-Backend$ ls -la
total 20
drwxr-xr-x  4 root root 4096 Oct  7 17:18 .
drwxr-xr-x  5 root root 4096 Oct  7 17:18 ..
drwxr-xr-x  7 root root 4096 Oct  7 17:18 .git
drwxr-xr-x 11 root root 4096 Oct  7 17:18 G6-PNC-STUDEND-SCORE-BACK
-rw-r--r--  1 root root   34 Oct  7 17:18 README.md
```

```
cd
/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK
ls -la
```

+ Make the web user own it & run composer install

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo chown -R www-data:www-data /var/www/Student-Score-Management-Backend
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ cd /var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK
composer install --no-dev --optimize-autoloader
The repository at "/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK" does not have the correct ownership and git refuses to use it:

fatal: detected dubious ownership in repository at '/var/www/Student-Score-Management-Backend'
To add an exception for this directory, call:

git config --global --add safe.directory /var/www/Student-Score-Management-Backend

Installing dependencies from lock file
Verifying lock file contents can be installed on current platform.
Package operations: 88 installs, 0 updates, 0 removals

In Filesystem.php line 267:

  /var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK/vendor does not exist and could not be created:

install [--prefer-source] [--prefer-dist] [--prefer-install PREFER-INSTALL] [--dry-run] [--download-only] [--dev] [--no-suggest] [--no-dev] [--no-security-blocking] [--n
o-blocking] [--no-autoloader] [--no-progress] [--no-install] [--audit] [--audit-format AUDIT-FORMAT] [-v|vv|vvv|--verbose] [-o|--optimize-autoloader] [-a|--classmap-auth
oritative] [--strict-psr-autoloader] [--apcu-autoloader] [--apcu-autoloader-prefix APCU-AUTOLOADER-PREFIX] [--ignore-platform-req IGNORE-PLATFORM-REQ] [--ignore-platform
-reqs] [--] [<packages>...]
```

```
sudo chown -R www-data:www-data
/var/www/Student-Score-Management-Backend
cd
/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK
composer install --no-dev --optimize-autoloader
```

- **Two separate issues & untangle them**

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ git config --global --add safe.directory /var/www/Student-Score-Management-Backend
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data composer install --no-dev --optimize-autoloader
Installing dependencies from lock file
Verifying lock file contents can be installed on current platform.
Cannot create cache directory /var/www/.cache/composer/repo/https---repo.packagist.org/, or directory is not writable. Proceeding without cache. See also cache-read-only
 config if your filesystem is read-only.
Package operations: 88 installs, 0 updates, 0 removals
Cannot create cache directory /var/www/.cache/composer/files/, or directory is not writable. Proceeding without cache. See also cache-read-only config if your filesystem
 is read-only.
  - Downloading doctrine/inflector (2.1.0)
  - Downloading doctrine/lexer (3.0.1)
  - Downloading dragonmantank/cron-expression (v3.6.0)
  - Downloading symfony/polyfill-mbstring (v1.38.2)
  - Downloading symfony/deprecation-contracts (v3.7.1)
  - Downloading symfony/http-foundation (v7.4.14)
  - Downloading fruitcake/php-cors (v1.4.0)
  - Downloading psr/log (3.0.2)
  - Downloading monolog/monolog (3.10.0)
  - Downloading symfony/polyfill-php80 (v1.37.0)
  - Downloading ralouphie/getallheaders (3.0.3)
  - Downloading psr/http-message (2.0)
  - Downloading psr/http-factory (1.1.0)
  - Downloading guzzlehttp/psr7 (2.12.5)
  - Downloading psr/http-client (1.0.3)
  - Downloading guzzlehttp/promises (2.5.1)
  - Downloading guzzlehttp/guzzle (7.14.1)
  - Downloading psr/cache (3.0.0)
  - Downloading firebase/php-jwt (v7.1.0)
  - Downloading google/auth (v1.52.0)
  - Downloading google/apiclient-services (v0.450.0)
  - Downloading google/apiclient (v2.19.4)
  - Downloading guzzlehttp/uri-template (v1.0.8)
  - Downloading symfony/polyfill-intl-normalizer (v1.38.0)
  - Downloading symfony/polyfill-intl-grapheme (v1.38.1)
  - Downloading symfony/polyfill-ctype (v1.37.0)
  - Downloading symfony/string (v7.4.13)
```

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ cat .env.example | grep -E "^DB_|^APP_"
APP_NAME=G6-PNC-Student-Score
APP_ENV=local                           # local | testing | production
APP_KEY=                                # Run `php artisan key:generate` after first clone
APP_DEBUG=true                          # Set to false in production
APP_URL=http://localhost                # Backend URL
APP_LOCALE=en
APP_FALLBACK_LOCALE=en
APP_FAKER_LOCALE=en_US
APP_MAINTENANCE_DRIVER=file             # file | database
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=student_score_db
DB_USERNAME=root
DB_PASSWORD=
```

  + **Git "dubious ownership"**

```
git config --global --add safe.directory
/var/www/Student-Score-Management-Backend
```

  + Composer can't write
    - **Option A — run composer as www-data** (keeps files owned by the web user, cleanest):

```
sudo -u www-data composer install --no-dev --optimize-autoloader
cat .env.example | grep -E "^DB_|^APP_"
```

+ Connect web server → MariaDB

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data cp .env.example .env
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan key:generate

  INFO  Application key set successfully.
```

```
sudo -u www-data cp .env.example .env
sudo -u www-data php artisan key:generate
```

- **Let's see what the migrations dictate before touching .env:**

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data ls database/migrations
0001_01_01_000000_create_users_table.php                    2026_07_08_000016_create_transcripts_table.php
0001_01_01_000001_create_cache_table.php                    2026_07_08_000018_cleanup_old_schema.php
0001_01_01_000002_create_jobs_table.php                     2026_07_08_000019_create_activity_logs_table.php
2026_07_07_010600_create_personal_access_tokens_table.php   2026_07_08_000021_create_report_card_details_table.php
2026_07_08_000001_create_departments_table.php              2026_07_08_000022_create_transcript_details_table.php
2026_07_08_000002_create_academic_years_table.php           2026_07_15_000001_add_subject_fields_to_subjects_table.php
2026_07_08_000003_create_roles_table.php                    2026_07_15_000001_create_grade_boundaries_table.php
2026_07_08_000004_create_permissions_table.php              2026_07_15_000002_create_subject_teacher_table.php
2026_07_08_000007_create_teachers_table.php                 2026_07_15_000004_add_room_to_classes.php
2026_07_08_000008_create_classes_table.php                  2026_07_16_000001_fix_student_id_number_column.php
2026_07_08_000008_create_subjects_table.php                 2026_07_16_000002_make_student_id_number_nullable.php
2026_07_08_000009_create_terms_table.php                    2026_07_16_000003_add_deleted_at_to_classes_table.php
2026_07_08_000010_create_students_table.php                 2026_07_16_012650_fix_enrollments_student_id_nullable.php
2026_07_08_000011_create_student_class_histories_table.php  2026_07_20_000001_add_google_refresh_token_to_users_table.php
2026_07_08_000011_create_subject_offerings_table.php        2026_07_21_000001_add_performance_indexes.php
2026_07_08_000012_create_student_subject_enrollments_table.php  2026_07_21_000002_add_teacher_id_to_classes.php
2026_07_08_000013_create_assessment_types_table.php         2026_07_22_000001_add_is_placeholder_to_students_table.php
2026_07_08_000013_create_scores_table.php                   2026_07_22_000002_backfill_is_placeholder_for_existing_imports.php
2026_07_08_000014_create_score_details_table.php            2026_07_23_000001_create_email_domain_rules_table.php
2026_07_08_000015_create_report_cards_table.php             2026_07_25_125757_add_bio_and_last_login_to_users_table.php
```

```
sudo -u www-data ls database/migrations
```

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ cat README.md | grep -iA4 "database\|migrate\|setup\|install" | head -50
| MySQL | >= 5.7 | Database |
| Laravel Sanctum | 4.3 | SPA Token Authentication |
| Eloquent ORM | - | Database ORM |
| Google API Client | 2.19 | Google Sheets OAuth2 |
| PHP Spreadsheet | 5.9 | Excel import/export |
| PDF Parser | * | PDF parsing |

--
├── database/
│   ├── migrations/                # 39 migration files
│   ├── seeders/                   # 10 seeders
│   │   ├── DatabaseSeeder.php
│   │   ├── AdminUserSeeder.php
│   │   ├── UsersSeeder.php
│   │   ├── TeacherSeeder.php
│   │   └── StudentSeeder.php
--
## Database Schema

### Core Tables

#### users
--
## Installation

```bash
git clone https://github.com/G6-PNC-STUDEND-SCORE/G6-PNC-STUDEND-SCORE-BACK.git
cd backend
composer install
cp .env.example .env
php artisan key:generate
php artisan migrate --seed
php artisan serve        # Runs at http://localhost:8000
```

Available Scripts
```

```
cat README.md | grep -iA4 "database\|migrate\|setup\|install" | head -50
```

3. Point `.env` at the DB (`DB_HOST=172.31.29.28`, etc.)

```
  GNU nano 8.7.1                    /var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK/.env *
# ──────────────────────────────────────────────────────────────
# PHP & Performance
# ──────────────────────────────────────────────────────────────
PHP_CLI_SERVER_WORKERS=4               # Dev server concurrency
BCRYPT_ROUNDS=12                       # Password hashing cost (10-14)

# ──────────────────────────────────────────────────────────────
# Logging
# ──────────────────────────────────────────────────────────────
LOG_CHANNEL=stack
LOG_STACK=single
LOG_DEPRECATIONS_CHANNEL=null
LOG_LEVEL=debug                        # debug | info | notice | warning | error | critical

# ──────────────────────────────────────────────────────────────
# Database (MySQL)
# ──────────────────────────────────────────────────────────────
DB_CONNECTION=mysql
DB_HOST=172.31.29.28
DB_PORT=3306
DB_DATABASE=score_db
DB_USERNAME=score_user
DB_PASSWORD=password

# ──────────────────────────────────────────────────────────────
# Session & Authentication
# ──────────────────────────────────────────────────────────────
SESSION_DRIVER=database
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was    M-F Next
```

DB_CONNECTION=mysql
DB_HOST=172.31.29.105 **(private IP of the database server)**
DB_PORT=3306
DB_DATABASE=score_db
DB_USERNAME=score_user
DB_PASSWORD=password

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate --pretend

  INFO  Nothing to migrate.
```

```
sudo -u www-data php artisan migrate --pretend
```

+ On the Database server

```
ubuntu@database-serer:~$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
ens5             UP             172.31.19.105/20 metric 100 fe80::8ff:edff:fe04:5a23/64
ubuntu@database-serer:~$
```

```
ip -br a
```

+ On Webserver

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ nc -zv 172.31.19.105 3306
Connection to 172.31.19.105 3306 port [tcp/mysql] succeeded!
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$
```

```
nc -zv 172.31.19.105 3306
```

+ Two things to fix on the **database server:**

```
ubuntu@database-serer:~$ sudo mariadb -e "SELECT user, host FROM mysql.user WHERE user='score_user';"
+------------+-------------+
| User       | Host        |
+------------+-------------+
| score_user | 172.31.29.28 |
+------------+-------------+
ubuntu@database-serer:~$ sudo mariadb -e "SHOW DATABASES;"
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| score_db           |
| sys                |
| wordpress_db       |
+--------------------+
ubuntu@database-serer:~$ sudo mariadb -e "SHOW DATABASES;"
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| score_db           |
| sys                |
| wordpress_db       |
+--------------------+
ubuntu@database-serer:~$
```

+ **Check in the database**

```
sudo mariadb -e "SELECT user, host FROM mysql.user WHERE user='score_user';"
sudo mariadb -e "SHOW DATABASES;"
sudo mariadb -e "SELECT user, host, authentication_string FROM mysql.user WHERE user='score_user';"
```

```
ubuntu@database-serer:~$ sudo mariadb -e "SELECT PASSWORD('password');"
+-------------------------------------------+
| PASSWORD('password')                      |
+-------------------------------------------+
| *2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19 |
+-------------------------------------------+
```

```
sudo -u www-data php artisan migrate --pretend
```

+ **On the web server:**

```
ubuntu@webserver:~$ sudo -u www-data php artisan migrate --pretend
Could not open input file: artisan
ubuntu@webserver:~$ cd /var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate --pretend

  INFO  Nothing to migrate.

ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate

  INFO  Nothing to migrate.

ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate:status

  Migration name ..... Batch / Status
  0001_01_01_000000_create_users_table ..... [1] Ran
  0001_01_01_000001_create_cache_table ..... [1] Ran
  0001_01_01_000002_create_jobs_table ..... [1] Ran
  2026_07_07_010600_create_personal_access_tokens_table ..... [1] Ran
  2026_07_08_000001_create_departments_table ..... [1] Ran
  2026_07_08_000002_create_academic_years_table ..... [1] Ran
  2026_07_08_000003_create_roles_table ..... [1] Ran
  2026_07_08_000004_create_permissions_table ..... [1] Ran
  2026_07_08_000007_create_teachers_table ..... [1] Ran
  2026_07_08_000008_create_classes_table ..... [1] Ran
  2026_07_08_000008_create_subjects_table ..... [1] Ran
  2026_07_08_000009_create_terms_table ..... [1] Ran
  2026_07_08_000010_create_students_table ..... [1] Ran
  2026_07_08_000011_create_student_class_histories_table ..... [1] Ran
  2026_07_08_000011_create_subject_offerings_table ..... [1] Ran
  2026_07_08_000012_create_student_subject_enrollments_table ..... [1] Ran
  2026_07_08_000013_create_assessment_types_table ..... [1] Ran
  2026_07_08_000013_create_scores_table ..... [1] Ran
  2026_07_08_000014_create_score_details_table ..... [1] Ran
  2026_07_08_000015_create_report_cards_table ..... [1] Ran
  2026_07_08_000016_create_transcripts_table ..... [1] Ran
  2026_07_08_000018_cleanup_old_schema ..... [1] Ran
```

```
sudo -u www-data php artisan migrate --pretend
```

+ **Migration SQL — no errors**

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate

  INFO  Nothing to migrate.

ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan migrate:status

  Migration name ..... Batch / Status
  0001_01_01_000000_create_users_table ..... [1] Ran
  0001_01_01_000001_create_cache_table ..... [1] Ran
  0001_01_01_000002_create_jobs_table ..... [1] Ran
  2026_07_07_010600_create_personal_access_tokens_table ..... [1] Ran
  2026_07_08_000001_create_departments_table ..... [1] Ran
  2026_07_08_000002_create_academic_years_table ..... [1] Ran
  2026_07_08_000003_create_roles_table ..... [1] Ran
  2026_07_08_000004_create_permissions_table ..... [1] Ran
  2026_07_08_000007_create_teachers_table ..... [1] Ran
  2026_07_08_000008_create_classes_table ..... [1] Ran
  2026_07_08_000008_create_subjects_table ..... [1] Ran
  2026_07_08_000009_create_terms_table ..... [1] Ran
  2026_07_08_000010_create_students_table ..... [1] Ran
  2026_07_08_000011_create_student_class_histories_table ..... [1] Ran
  2026_07_08_000011_create_subject_offerings_table ..... [1] Ran
  2026_07_08_000012_create_student_subject_enrollments_table ..... [1] Ran
  2026_07_08_000013_create_assessment_types_table ..... [1] Ran
  2026_07_08_000013_create_scores_table ..... [1] Ran
  2026_07_08_000014_create_score_details_table ..... [1] Ran
  2026_07_08_000015_create_report_cards_table ..... [1] Ran
  2026_07_08_000016_create_transcripts_table ..... [1] Ran
  2026_07_08_000018_cleanup_old_schema ..... [1] Ran
```

```
sudo -u www-data php artisan migrate
sudo -u www-data php artisan migrate:status
```

+ **Check in the database**

```
ubuntu@database-serer:~$ sudo mariadb -e "SHOW TABLES FROM score_db;"
+-----------------------------+
| Tables_in_score_db          |
+-----------------------------+
| academic_years              |
| activity_logs               |
| assessment_types            |
| cache                       |
| cache_locks                 |
| classes                     |
| departments                 |
| email_domain_rules          |
| failed_jobs                 |
| generations                 |
| grade_boundaries            |
| job_batches                 |
| jobs                        |
| migrations                  |
| password_reset_tokens       |
| permissions                 |
| personal_access_tokens      |
| report_card_details         |
| report_cards                |
| role_permission             |
| roles                       |
| score_details               |
| scores                      |
| sessions                    |
| student_class_histories     |
| student_subject_enrollments |
| students                    |
| subject_offerings           |
| subject_teacher             |
| subject_term                |
| subjects                    |
```

`sudo mariadb -e "SHOW TABLES FROM score_db;"`

+ **From webserver**

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ mariadb -h 172.31.19.105 -u score_user -p'password' score_db -e "SHOW TABLES;"
+-----------------------------+
| Tables_in_score_db          |
+-----------------------------+
| academic_years              |
| activity_logs               |
| assessment_types            |
| cache                       |
| cache_locks                 |
| classes                     |
| departments                 |
| email_domain_rules          |
| failed_jobs                 |
| generations                 |
| grade_boundaries            |
| job_batches                 |
| jobs                        |
| migrations                  |
| password_reset_tokens       |
| permissions                 |
| personal_access_tokens      |
| report_card_details         |
| report_cards                |
| role_permission             |
| roles                       |
| score_details               |
| scores                      |
| sessions                    |
| student_class_histories     |
| student_subject_enrollments |
| students                    |
| subject_offerings           |
| subject_teacher             |
| subject_term                |
| subjects                    |
| teachers                    |
```

`mariadb -h 172.31.19.105 -u score_user -p score_db -e "SHOW TABLES;"`

+ Seed the data (webserver)

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan db:seed

  INFO  Seeding database.

  Database\Seeders\UsersSeeder ..... RUNNING
  Database\Seeders\UsersSeeder ..... 1,292 ms DONE

  Database\Seeders\EmailDomainRuleSeeder ..... RUNNING
  Database\Seeders\EmailDomainRuleSeeder ..... 13 ms DONE

  Database\Seeders\PermissionSeeder ..... RUNNING
  Database\Seeders\PermissionSeeder ..... 372 ms DONE

  Database\Seeders\TeacherSeeder ..... RUNNING
  Database\Seeders\TeacherSeeder ..... 124 ms DONE

  Database\Seeders\SubjectSeeder ..... RUNNING
  Database\Seeders\SubjectSeeder ..... 194 ms DONE

  Database\Seeders\AssessmentTypeSeeder ..... RUNNING
  Database\Seeders\AssessmentTypeSeeder ..... 76 ms DONE

  Database\Seeders\StudentSeeder ..... RUNNING
  Database\Seeders\StudentSeeder ..... 232 ms DONE

  Database\Seeders\SubjectTermSeeder ..... RUNNING
Subject-Term pivot table seeded successfully!
  - Inserted: 23 new entries
  - Skipped (already exist): 0 entries

Subject-Term Assignment Summary:
------------------------------------------------------------
  Term 1: Logic, BCU, MS Office, English, PL, Design
  Term 2: Web design, Algorithms, Git, English for IT, PL
  Term 3: PHP, Database, Javascript, QA, English, English for IT, PL
  Term 4: Laravel, Node.js, Vue.js, OOP, Typescript
```

```
sudo -u www-data php artisan db:seed
```

+ **Check what seeders exist first:**

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data php artisan db:show  # or just run db:seed --force


  MariaDB ..... 11.8.6
  Connection ..... mysql
  Database ..... score_db
  Host ..... 172.31.19.105
  Port ..... 3306
  Username ..... score_user
  URL .....
  Open Connections .....
  Tables ..... 36
  Total Size ..... 2.09 MB

  Schema / Table ..... Size
  score_db / academic_years ..... 32.00 KB
  score_db / activity_logs ..... 96.00 KB
  score_db / assessment_types ..... 48.00 KB
  score_db / cache ..... 16.00 KB
  score_db / cache_locks ..... 16.00 KB
  score_db / classes ..... 48.00 KB
  score_db / departments ..... 16.00 KB
  score_db / email_domain_rules ..... 48.00 KB
  score_db / failed_jobs ..... 32.00 KB
  score_db / generations ..... 48.00 KB
  score_db / grade_boundaries ..... 32.00 KB
  score_db / jobs ..... 32.00 KB
  score_db / job_batches ..... 16.00 KB
  score_db / migrations ..... 16.00 KB
  score_db / password_reset_tokens ..... 16.00 KB
  score_db / permissions ..... 48.00 KB
  score_db / personal_access_tokens ..... 64.00 KB
  score_db / report_cards ..... 144.00 KB
  score_db / report_card_details ..... 64.00 KB
```

```
sudo -u www-data php artisan db:show  # or just run db:seed --force
```

+ **Run the server to test:**

```
sudo -u www-data php artisan serve --host=127.0.0.1 --port=8000
```

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ sudo -u www-data nohup php artisan serve --host=0.0.0.0 --port=8000 > /tmp/laravel.
log 2>&1 &
curl -s -o /dev/null -w "%{http_code}\n" http://localhost:8000
[3] 100398
[2]   Exit 1                  sudo -u www-data nohup php artisan serve --host=0.0.0.0 --port=8000 > /tmp/laravel.log 2>&1
200
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$
```

```
sudo -u www-data nohup php artisan serve --host=127.0.0.1
--port=8000 > /tmp/laravel.log 2>&1 & curl -s -o /dev/null -w
"%{http_code}\n" http://localhost:8000
```

```
  GNU nano 8.7.1                                  /etc/apache2/sites-available/laravel.conf *
<VirtualHost *:80>
    ServerName studentscoremanagement.narong.world
    ServerAlias www.studentscoremanagement.narong.world
    ProxyPreserveHost On
    ProxyPass / http://127.0.0.1:8000/
    ProxyPassReverse / http://127.0.0.1:8000/
</VirtualHost>
```

4. Clone repo (Frontend)

```
ubuntu@webserver:/var/www/Student-Score-Management-Backend/G6-PNC-STUDEND-SCORE-BACK$ cd /var/www/
ubuntu@webserver:/var/www$ sudo git clone https://github.com/riengnarong-alt/Student-Score-Management-frontend.git
Cloning into 'Student-Score-Management-frontend'...
remote: Enumerating objects: 172, done.
remote: Counting objects: 100% (172/172), done.
remote: Compressing objects: 100% (154/154), done.
remote: Total 172 (delta 9), reused 172 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (172/172), 542.90 KiB | 21.72 MiB/s, done.
Resolving deltas: 100% (9/9), done.
ubuntu@webserver:/var/www$
```

cd /var/www/
sudo git clone https://github.com/riengnarong-alt/Student-Score-Management-frontend.git

5. nested folder

```
ubuntu@webserver:/var/www$ ls /var/www/Student-Score-Management-frontend/
G6-PNC-STUDEND-SCORE-FRONT
ubuntu@webserver:/var/www$ ls /var/www/Student-Score-Management-frontend/
G6-PNC-STUDEND-SCORE-FRONT
ubuntu@webserver:/var/www$ ls /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/
README.md  env.d.ts  eslint.config.ts  index.html  package-lock.json  package.json  public  src  tsconfig.app.json  tsconfig.json  tsconfig.node.json  vite.config.ts
ubuntu@webserver:/var/www$ cat /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/package.json
{
  "name": "student-score-management-system",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "run-p type-check \"build-only {@}\" --",
    "preview": "vite preview",
    "build-only": "vite build",
    "type-check": "vue-tsc --build",
    "lint": "run-s \"lint:*\"",
    "lint:oxlint": "oxlint . --fix",
    "lint:eslint": "eslint . --fix --cache",
    "format": "oxfmt src/"
  },
  "dependencies": {
    "@lucide/vue": "^1.24.0",
    "@popperjs/core": "^2.11.8",
    "axios": "^1.18.1",
    "bootstrap": "^5.3.8",
    "bootstrap-icons": "^1.13.1",
    "chart.js": "^4.5.1",
    "echarts": "^6.1.0",
    "i18n": "^0.15.3",
    "jspdf": "^4.2.1",
    "jspdf-autotable": "^5.0.8",
    "lucide-vue-next": "^1.0.0",
    "pdfjs-dist": "^6.1.200",
```

ls /var/www/Student-Score-Management-frontend/
cat /var/www/Student-Score-Management-frontend/package.json
ls /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/
cat /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/package.json

+ **Check Node version**

```
ubuntu@webserver:/var/www$ node -v
v22.23.3
ubuntu@webserver:/var/www$
```

6. Build the frontend

```
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ sudo npm run build
npm notice run student-score-management-system@0.0.0 build
npm notice run run-p type-check "build-only {@}" --
npm notice run student-score-management-system@0.0.0 type-check
npm notice run vue-tsc --build
npm notice run student-score-management-system@0.0.0 build-only
npm notice run vite build
vite v8.1.5 building client environment for production...
✓ 4529 modules transformed.
computing gzip size...
dist/index.html                                  2.54 kB | gzip:   1.04 kB
dist/assets/bootstrap-icons-mSm7cUeB.woff2     134.04 kB
dist/assets/bootstrap-icons-BeopsB42.woff      180.28 kB
dist/assets/pnc-logo-C_DyKTD4.png              208.27 kB
dist/assets/pdf.worker.min-DEtVeC4l.mjs      1,255.06 kB
dist/assets/LoadingState-DkPUBRaw.css            0.04 kB | gzip:   0.06 kB
dist/assets/SettingsPage-DxX5vh5S.css            0.37 kB | gzip:   0.23 kB
dist/assets/DataTable-DSP28QPB.css               2.19 kB | gzip:   0.70 kB
dist/assets/TermSubjectsView-pFrK01T-.css        5.02 kB | gzip:   1.36 kB
dist/assets/ClassPage-DdsTnIjt.css               6.33 kB | gzip:   1.71 kB
dist/assets/StudentScoresPage-CHPV18Rq.css       6.43 kB | gzip:   1.73 kB
dist/assets/ForgotPasswordPage-BaIxkaN4.css      7.11 kB | gzip:   2.05 kB
dist/assets/ResetPasswordPage-CneB0pud.css       7.12 kB | gzip:   2.06 kB
dist/assets/TeacherPage-uaZT3m0-.css             7.95 kB | gzip:   2.02 kB
dist/assets/LoginPage-Hfse5J5-.css               9.01 kB | gzip:   2.43 kB
dist/assets/StudentTranscriptPage-4t_itpZQ.css   9.02 kB | gzip:   2.21 kB
dist/assets/UsersPage-xBRkFN6D.css               9.96 kB | gzip:   2.43 kB
dist/assets/StudentPortalPage-C-2Y0Gr8.css      12.30 kB | gzip:   2.85 kB
dist/assets/ActivityLogPage-Dso3-1_J.css        13.07 kB | gzip:   2.86 kB
dist/assets/ScorePage-5tZspa8k.css              13.61 kB | gzip:   2.61 kB
dist/assets/SubjectPage-DfBCT91Z.css            17.05 kB | gzip:   3.51 kB
dist/assets/ReportPage-BRjLM3Gi.css             17.06 kB | gzip:   3.37 kB
dist/assets/RolesPage-lPXExa8l.css              17.63 kB | gzip:   3.23 kB
dist/assets/MainLayout-BDhgUL3m.css             18.07 kB | gzip:   3.40 kB
dist/assets/UserProfile-UjKdjNsV.css            20.30 kB | gzip:   3.74 kB
dist/assets/DashboardView-C0KJSTW6.css          23.45 kB | gzip:   4.50 kB
```

```
cd
/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/
sudo chown -R $(whoami):$(whoami)
/var/www/Student-Score-Management-frontend/
npm install
npm run build
```

7. Create the .env & rebuild

```
 GNU nano 8.7.1                    /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/.env *
VITE_API_BASE_URL=/api
```

```
sudo nano
/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/.env
```

Paste this single line, save (Ctrl+O, Enter, Ctrl+X):
VITE_API_BASE_URL=/api

- **rebuild:**

```
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ sudo npx vite build
npm notice run student-score-management-system@0.0.0 npx
npm notice run 'vite' build
vite v8.1.5 building client environment for production...
✓ 4529 modules transformed.
computing gzip size...
dist/index.html                                  2.54 kB │ gzip:   1.04 kB
dist/assets/bootstrap-icons-mSm7cUeB.woff2     134.04 kB
dist/assets/bootstrap-icons-BeopsB42.woff      180.28 kB
dist/assets/pnc-logo-C_DyKTD4.png              208.27 kB
dist/assets/pdf.worker.min-DEtVeC4l.mjs      1,255.06 kB
dist/assets/LoadingState-DkPUBRaw.css            0.04 kB │ gzip:   0.06 kB
dist/assets/SettingsPage-DxX5vh5S.css            0.37 kB │ gzip:   0.23 kB
dist/assets/DataTable-DSP28QPB.css               2.19 kB │ gzip:   0.70 kB
dist/assets/TermSubjectsView-pFrK01T-.css        5.02 kB │ gzip:   1.36 kB
dist/assets/ClassPage-DdsTnIjt.css               6.33 kB │ gzip:   1.71 kB
dist/assets/StudentScoresPage-CHPV18Rq.css       6.43 kB │ gzip:   1.73 kB
dist/assets/ForgotPasswordPage-BaIxkaN4.css      7.11 kB │ gzip:   2.05 kB
dist/assets/ResetPasswordPage-CneB0pud.css       7.12 kB │ gzip:   2.06 kB
dist/assets/TeacherPage-uaZT3m0-.css             7.95 kB │ gzip:   2.02 kB
dist/assets/LoginPage-Hfse5J5-.css               9.01 kB │ gzip:   2.43 kB
dist/assets/StudentTranscriptPage-4t_itpZQ.css   9.02 kB │ gzip:   2.21 kB
dist/assets/UsersPage-xBRkFN6D.css               9.96 kB │ gzip:   2.43 kB
dist/assets/StudentPortalPage-C-2Y0Gr8.css      12.30 kB │ gzip:   2.85 kB
dist/assets/ActivityLogPage-Dso3-1_J.css        13.07 kB │ gzip:   2.86 kB
dist/assets/ScorePage-5tZspa8k.css              13.61 kB │ gzip:   2.61 kB
dist/assets/SubjectPage-DfBCT91Z.css            17.05 kB │ gzip:   3.51 kB
dist/assets/ReportPage-BRjLM3Gi.css             17.06 kB │ gzip:   3.37 kB
dist/assets/RolesPage-lPXExa8l.css              17.63 kB │ gzip:   3.23 kB
dist/assets/MainLayout-BDhgUL3m.css             18.07 kB │ gzip:   3.40 kB
dist/assets/UserProfile-UjKdjNsV.css            20.30 kB │ gzip:   3.74 kB
dist/assets/DashboardView-C0KJSTW6.css          23.45 kB │ gzip:   4.50 kB
dist/assets/index-iYt3__sT.css                  25.75 kB │ gzip:   5.14 kB
dist/assets/StudentPage-DgaNaveU.css            31.65 kB │ gzip:   5.25 kB
dist/assets/ScoreSheetView-dTJvaAUe.css         61.27 kB │ gzip:   9.94 kB
```

npx vite build

8.enable proxy + restart Apache

```
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ sudo a2enmod proxy proxy_http proxy_balancer
sudo systemctl restart apache2
Module proxy already enabled
Considering dependency proxy for proxy_http:
Module proxy already enabled
Module proxy_http already enabled
Considering dependency proxy for proxy_balancer:
Module proxy already enabled
Considering dependency alias for proxy_balancer:
Module alias already enabled
Considering dependency slotmem_shm for proxy_balancer:
Enabling module slotmem_shm.
Enabling module proxy_balancer.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ sudo systemctl restart apache2
```

sudo a2enmod proxy proxy_http proxy_balancer
sudo systemctl restart apache2

9.Create the vhost

sudo nano /etc/apache2/sites-available/studentscoremanagement.narong.world.conf

```
  GNU nano 8.7.1                                  /etc/apache2/sites-available/studentscoremanagement.narong.world.conf *
<VirtualHost *:80>
    ServerName studentscoremanagement.narong.world
    ServerAdmin webmaster@narong.world
    DocumentRoot /var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/dist

    <Directory "/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT/dist">
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ProxyPreserveHost On
    ProxyPass /api http://127.0.0.1:8000/api
    ProxyPassReverse /api http://127.0.0.1:8000/api

    ErrorLog ${APACHE_LOG_DIR}/ssm-error.log
    CustomLog ${APACHE_LOG_DIR}/ssm-access.log combined
</VirtualHost>
```

```
sudo a2ensite studentscoremanagement.narong.world
sudo systemctl reload apache2
```

10. Create Certificate + HTTPS

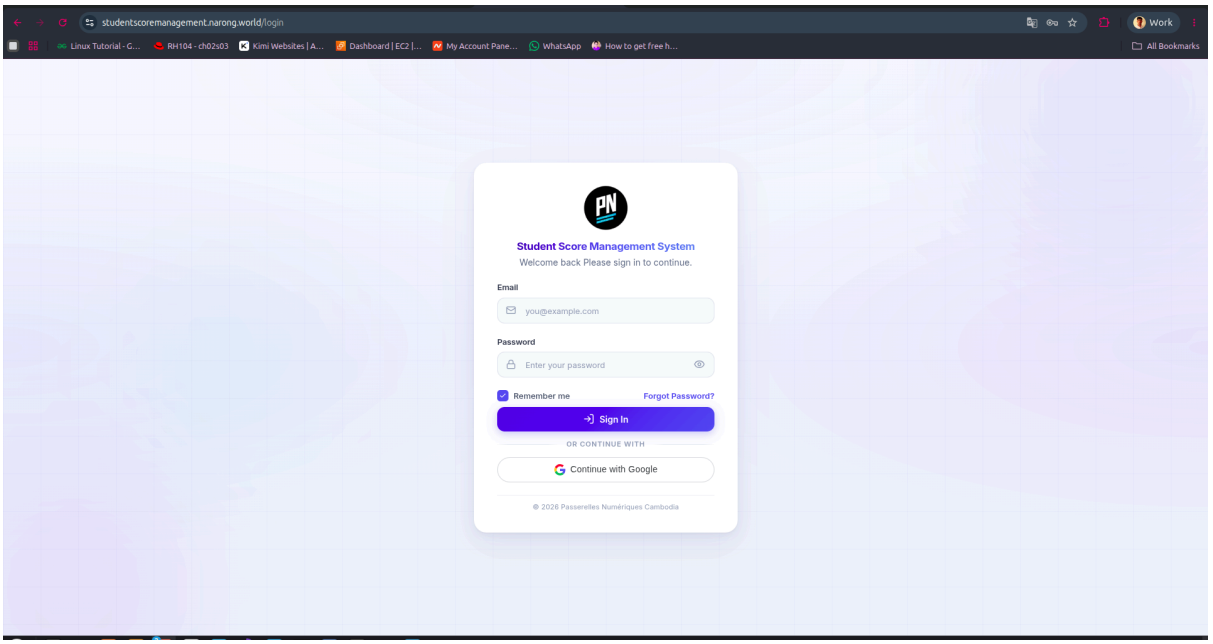
```
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ sudo certbot --apache -d studentscoremanagement.narong.world
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for studentscoremanagement.narong.world

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/studentscoremanagement.narong.world/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/studentscoremanagement.narong.world/privkey.pem
This certificate expires on 2027-01-06.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for studentscoremanagement.narong.world to /etc/apache2/sites-available/studentscoremanagement.narong.world-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://studentscoremanagement.narong.world

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@webserver:/var/www/Student-Score-Management-frontend/G6-PNC-STUDEND-SCORE-FRONT$ curl -I https://studentscoremanagement.narong.world/
HTTP/1.1 200 OK
Date: Thu, 08 Oct 2026 02:40:27 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Thu, 08 Oct 2026 02:21:55 GMT
ETag: "9f4-65d4ae53aec85"
Accept-Ranges: bytes
Content-Length: 2548
Vary: Accept-Encoding
Content-Type: text/html
```

```
sudo certbot --apache -d studentscoremanagement.narong.world
curl -I https://studentscoremanagement.narong.world/
```



+ Set API

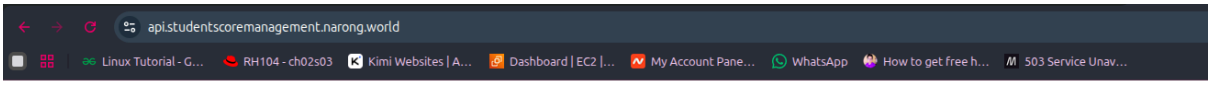
```
  GNU nano 8.7.1                                  /etc/apache2/sites-available/api.studentscoremanagement.narong.world.conf *
<VirtualHost *:80>
    ServerName api.studentscoremanagement.narong.world

    DocumentRoot /var/www/score-api/public

    <Directory /var/www/score-api/public>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    <FilesMatch \.php$>
        SetHandler "proxy:unix:/run/php/php8.5-fpm.sock|fcgi://localhost"
    </FilesMatch>

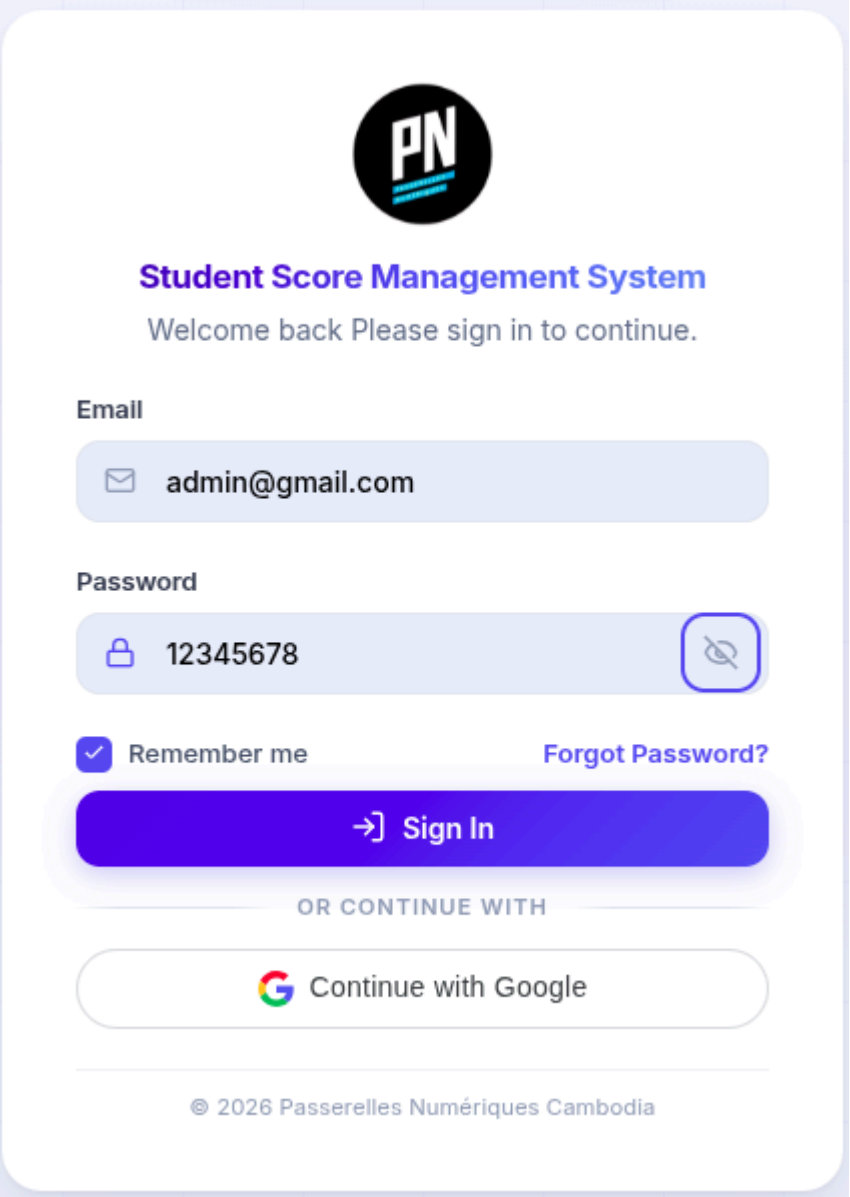
    ErrorLog ${APACHE_LOG_DIR}/api-error.log
    CustomLog ${APACHE_LOG_DIR}/api-access.log combined
</VirtualHost>
```



PNC Student Score Management System APIs is running.

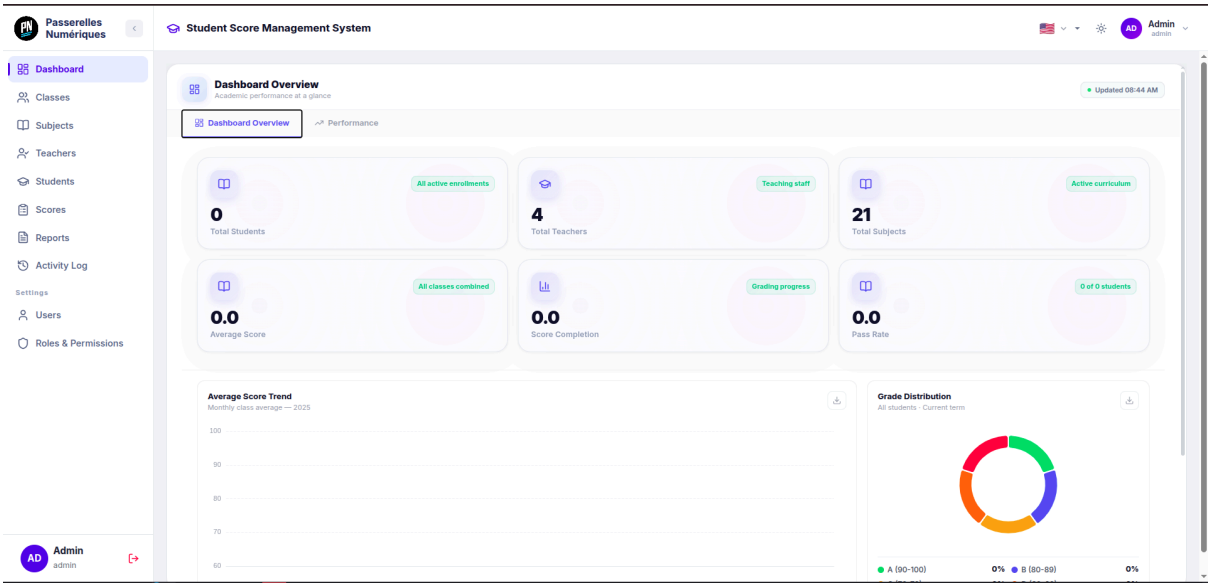
11. Login into Student Score Management System website

https://studentscoremanagement.narong.world/login



+ **Log in as an Admin**
- **Email:** admin@gmail.com
- **Password:** 12345678

+ The whole dashboard



Done for Student Score Management System

Hosting WordPress

(Web-server)

1. Create Database User Privileges (Database Server)

```
ubuntu@database-serer:~$ sudo mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 2124
Server version: 11.8.6-MariaDB-5ubuntu0.1 from Ubuntu -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE USER 'wp_user'@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON wordpress_db.* TO 'wp_user'@'%';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

```
CREATE DATABASE IF NOT EXISTS wordpress_db CHARACTER SET utf8mb4;
CREATE USER 'wp_user'@'172.31.29.28' IDENTIFIED BY
'CHANGE_ME_strong_password';
GRANT SELECT, INSERT, UPDATE, DELETE, CREATE, ALTER, DROP, INDEX ON
wordpress_db.* TO 'wp_user'@'172.31.29.28';
FLUSH PRIVILEGES;
```

+ **Verify User Creation**

```
MariaDB [(none)]> select user, host from mysql.user where user= 'wp_user';
+---------+------+
| User    | Host |
+---------+------+
| wp_user | %    |
+---------+------+
1 row in set (0.001 sec)

MariaDB [(none)]>
```

```
SELECT user, host FROM mysql.user WHERE user = 'wp_user';
```

(Web-server)

2. Create the Web Directory

```
ubuntu@webserver:~$ sudo mkdir -p /var/www/site
ubuntu@webserver:~$ ls /var/www/site/
ubuntu@webserver:~$ ls -la /var/www/site/
total 8
drwxr-xr-x 2 root root 4096 Oct  9 03:25 .
drwxr-xr-x 7 root root 4096 Oct  9 03:25 ..
ubuntu@webserver:~$
```

```
sudo mkdir -p /var/www/site
```

3. Download and Extract WordPress

```
wordpress/wp-includes/view-config.php
wordpress/wp-includes/view-transitions.php
wordpress/wp-includes/widgets/
wordpress/wp-includes/widgets/class-wp-nav-menu-widget.php
wordpress/wp-includes/widgets/class-wp-widget-archives.php
wordpress/wp-includes/widgets/class-wp-widget-block.php
wordpress/wp-includes/widgets/class-wp-widget-calendar.php
wordpress/wp-includes/widgets/class-wp-widget-categories.php
wordpress/wp-includes/widgets/class-wp-widget-custom-html.php
wordpress/wp-includes/widgets/class-wp-widget-links.php
wordpress/wp-includes/widgets/class-wp-widget-media-audio.php
wordpress/wp-includes/widgets/class-wp-widget-media-gallery.php
wordpress/wp-includes/widgets/class-wp-widget-media-image.php
wordpress/wp-includes/widgets/class-wp-widget-media-video.php
wordpress/wp-includes/widgets/class-wp-widget-media.php
wordpress/wp-includes/widgets/class-wp-widget-meta.php
wordpress/wp-includes/widgets/class-wp-widget-pages.php
wordpress/wp-includes/widgets/class-wp-widget-recent-comments.php
wordpress/wp-includes/widgets/class-wp-widget-recent-posts.php
wordpress/wp-includes/widgets/class-wp-widget-rss.php
wordpress/wp-includes/widgets/class-wp-widget-search.php
wordpress/wp-includes/widgets/class-wp-widget-tag-cloud.php
wordpress/wp-includes/widgets/class-wp-widget-text.php
wordpress/wp-includes/widgets.php
wordpress/wp-includes/wp-db.php
wordpress/wp-includes/wp-diff.php
wordpress/wp-links-opml.php
wordpress/wp-load.php
wordpress/wp-login.php
wordpress/wp-mail.php
wordpress/wp-settings.php
wordpress/wp-signup.php
wordpress/wp-trackback.php
wordpress/xmlrpc.php
ubuntu@webserver:/tmp$ sudo cp -a wordpress/. /var/www/site/
ubuntu@webserver:/tmp$ rm -rf wordpress latest.tar.gz
```

```
cd /tmp
curl -O https://wordpress.org/latest.tar.gz
tar -xvf latest.tar.gz
sudo cp -a wordpress/. /var/www/site/
rm -rf wordpress latest.tar.gz
ls -la /var/www/site/
```

4. Set Permissions and Ownership

```
ubuntu@webserver:/tmp$ sudo chown -R www-data:www-data /var/www/site
ubuntu@webserver:/tmp$ sudo chmod -R 755 /var/www/site
ubuntu@webserver:/tmp$
```

```
sudo chown -R www-data:www-data /var/www/site
sudo find /var/www/site -type d -exec chmod 755 {} \;
sudo find /var/www/site -type f -exec chmod 644 {} \;
```

5. Configure Apache Virtual Host

```
  GNU nano 8.7.1                                  /etc/apache2/sites-available/site.conf *
<VirtualHost *:80>
    ServerAdmin admin@gmail.com
    ServerName wordpress.narong.world
    DocumentRoot /var/www/site

    <Directory /var/www/site>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/site_error.log
    CustomLog ${APACHE_LOG_DIR}/site_access.log combined
</VirtualHost>
```

```
sudo nano /etc/apache2/sites-available/site.conf
```

6. Enable Site and Restart Apache

```
ubuntu@webserver:/tmp$ sudo nano /etc/apache2/sites-available/site.conf
ubuntu@webserver:/tmp$ sudo a2ensite site.conf
Enabling site site.
To activate the new configuration, you need to run:
  systemctl reload apache2
ubuntu@webserver:/tmp$
sudo a2enmod rewrite
Module rewrite already enabled
ubuntu@webserver:/tmp$
sudo apache2ctl configtest
Syntax OK
ubuntu@webserver:/tmp$ sudo systemctl restart apache2
```

+ **Enable the site configuration**

```
sudo a2ensite site.conf
```

+ **Enable the rewrite module for WordPress permalinks**

```
sudo a2enmod rewrite
```

+ **Test configuration syntax**

```
sudo apache2ctl configtest
```

+ **Restart Apache to apply changes**

```
sudo systemctl restart apache2
```

7. Install PHP and Apache Module

```
ubuntu@webserver:/tmp$ sudo apt install php libapache2-mod-php php-mysql -y
php-mysql is already the newest version (2:8.5+99ubuntu1).
Installing:
  libapache2-mod-php  php

Installing dependencies:
  libapache2-mod-php8.5  php8.5

Suggested packages:
  php-pear

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 52
  Download size: 3087 kB
  Space needed: 10.2 MB / 10.2 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 libapache2-mod-php8.5 amd64 8.5.4-0ubuntu1.3 [3069 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libapache2-mod-php all 2:8.5+99ubuntu1 [4316 B]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 php8.5 all 8.5.4-0ubuntu1.3 [9168 B]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 php all 2:8.5+99ubuntu1 [4164 B]
Fetched 3087 kB in 0s (58.0 MB/s)
Selecting previously unselected package libapache2-mod-php8.5.
(Reading database ... 132775 files and directories currently installed.)
Preparing to unpack .../libapache2-mod-php8.5_8.5.4-0ubuntu1.3_amd64v3.deb ...
Unpacking libapache2-mod-php8.5 (8.5.4-0ubuntu1.3) ...
Selecting previously unselected package libapache2-mod-php.
Preparing to unpack .../libapache2-mod-php_2%3a8.5+99ubuntu1_all.deb ...
Unpacking libapache2-mod-php (2:8.5+99ubuntu1) ...
Selecting previously unselected package php8.5.
Preparing to unpack .../php8.5_8.5.4-0ubuntu1.3_all.deb ...
Unpacking php8.5 (8.5.4-0ubuntu1.3) ...
Selecting previously unselected package php.
Preparing to unpack .../php_2%3a8.5+99ubuntu1_all.deb ...
Unpacking php (2:8.5+99ubuntu1) ...
Setting up libapache2-mod-php8.5 (8.5.4-0ubuntu1.3) ...
Creating config file /etc/php/8.5/apache2/php.ini with new version
```

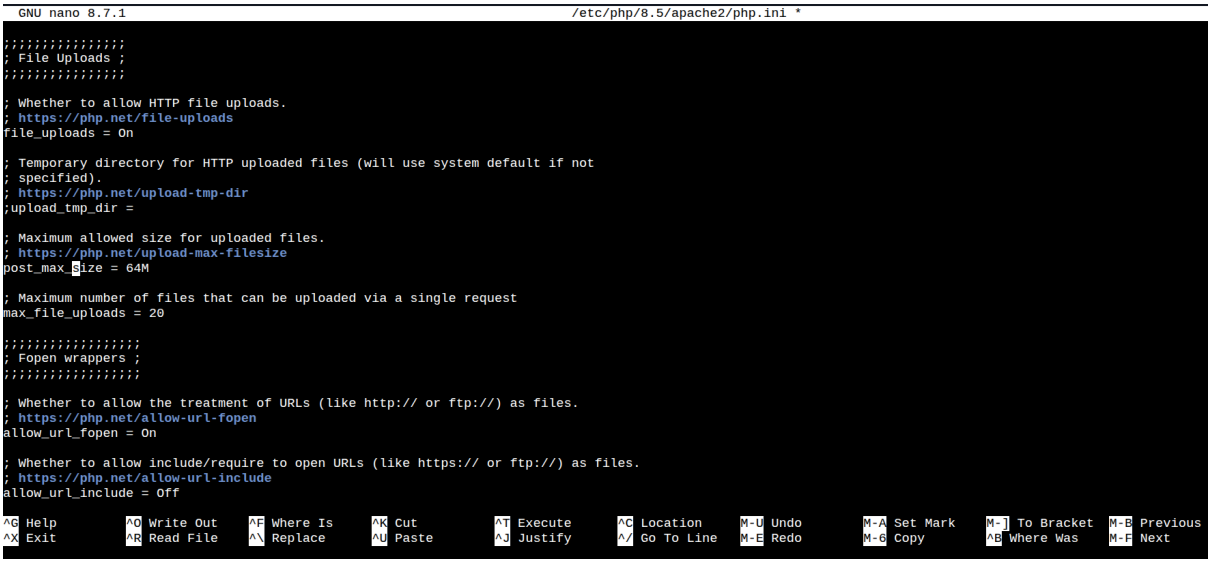
```
sudo apt update
sudo apt install php libapache2-mod-php php-mysql -y
```

+ Enable PHP Module and Restart Apache

```
ubuntu@webserver:/tmp$ sudo a2enmod php*
Considering dependency mpm_prefork for php8.5:
Considering conflict mpm_event for mpm_prefork:
Considering conflict mpm_worker for mpm_prefork:
Module mpm_prefork already enabled
Considering conflict php5 for php8.5:
Module php8.5 already enabled
ubuntu@webserver:/tmp$ sudo systemctl restart apache2
```

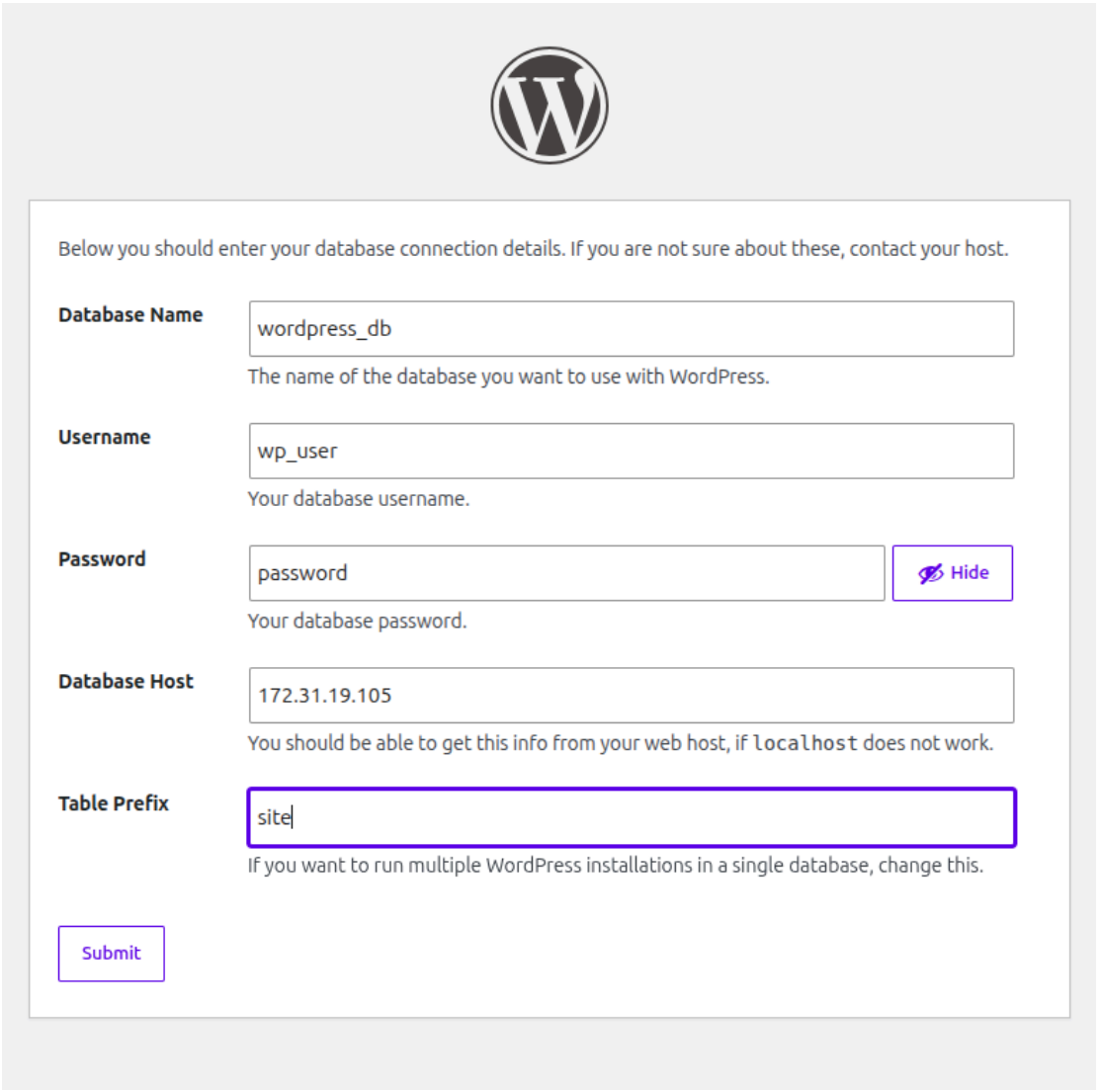
```
# Not needed: libapache2-mod-php already enables PHP (check:
apache2ctl -M | grep php)
sudo systemctl restart apache2
```

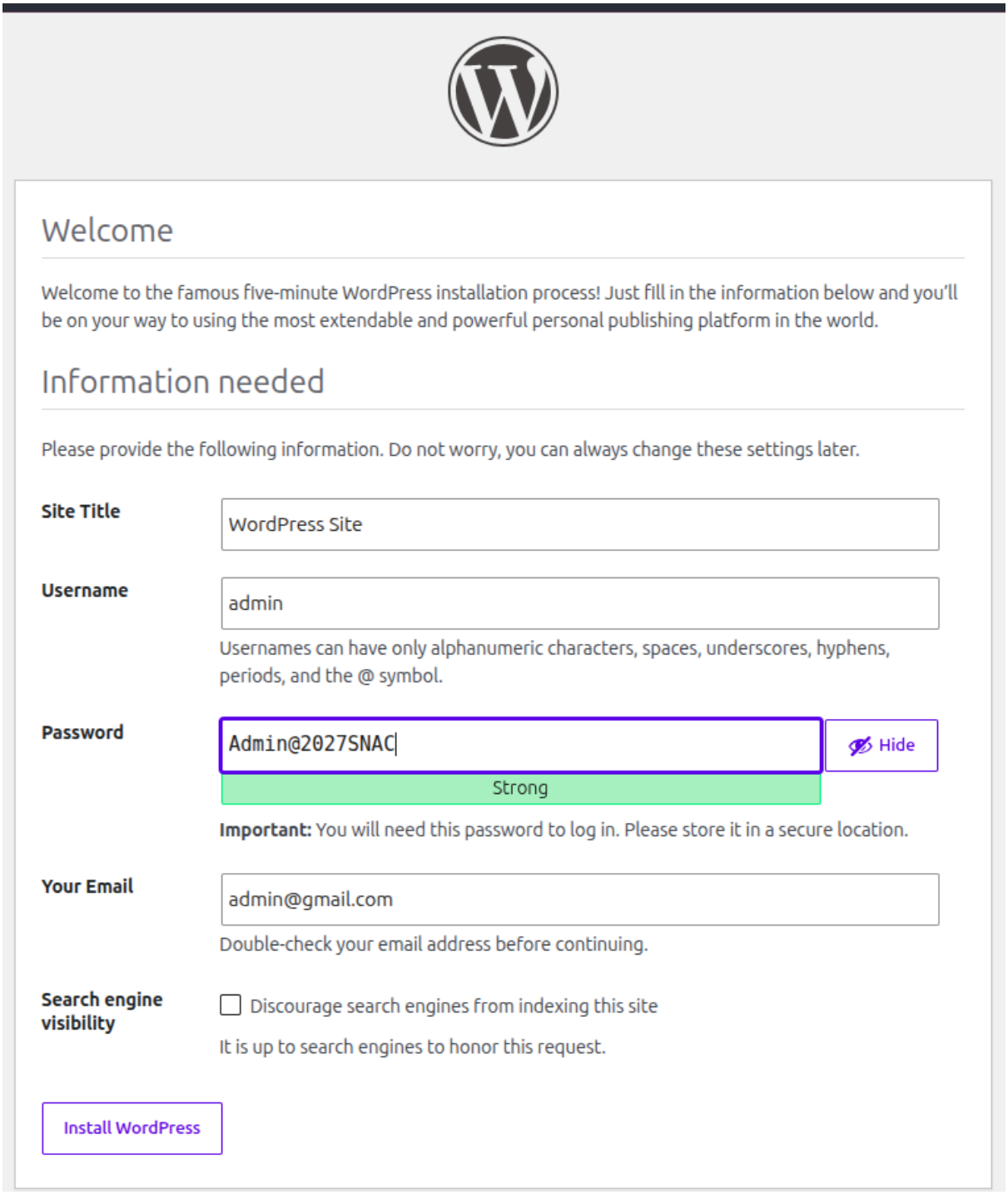
+ Increase PHP upload Limit (Optional)




```
sudo nano /etc/php/8.5/apache2/php.ini
# Set:  upload_max_filesize = 64M
# Set:  post_max_size = 64M
# Then: sudo systemctl restart apache2
```

+ Login on Browser





Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title WordPress Site

Username admin

Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password Admin@2027SNAC Hide

Strong

Important: You will need this password to log in. Please store it in a secure location.

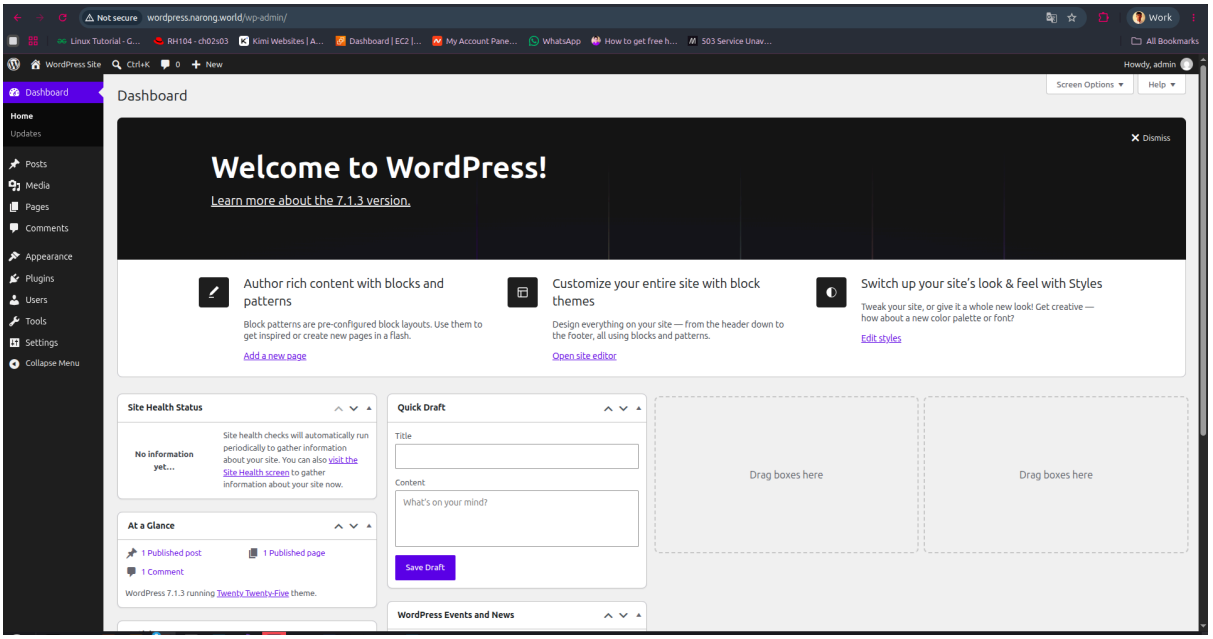
Your Email admin@gmail.com

Double-check your email address before continuing.

Search engine visibility Discourage search engines from indexing this site

It is up to search engines to honor this request.

Install WordPress



8. Install Certbot

```
ubuntu@webserver:/tmp$ sudo apt install certbot python3-certbot-apache -y

certbot is already the newest version (4.0.0-4).
python3-certbot-apache is already the newest version (4.0.0-3).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 52
```

```
sudo apt update
sudo apt install certbot python3-certbot-apache -y
```

9. Obtain & Configure SSL Certificate

```
ubuntu@webserver:/tmp$ sudo certbot --apache -d wordpress.narong.world
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for wordpress.narong.world

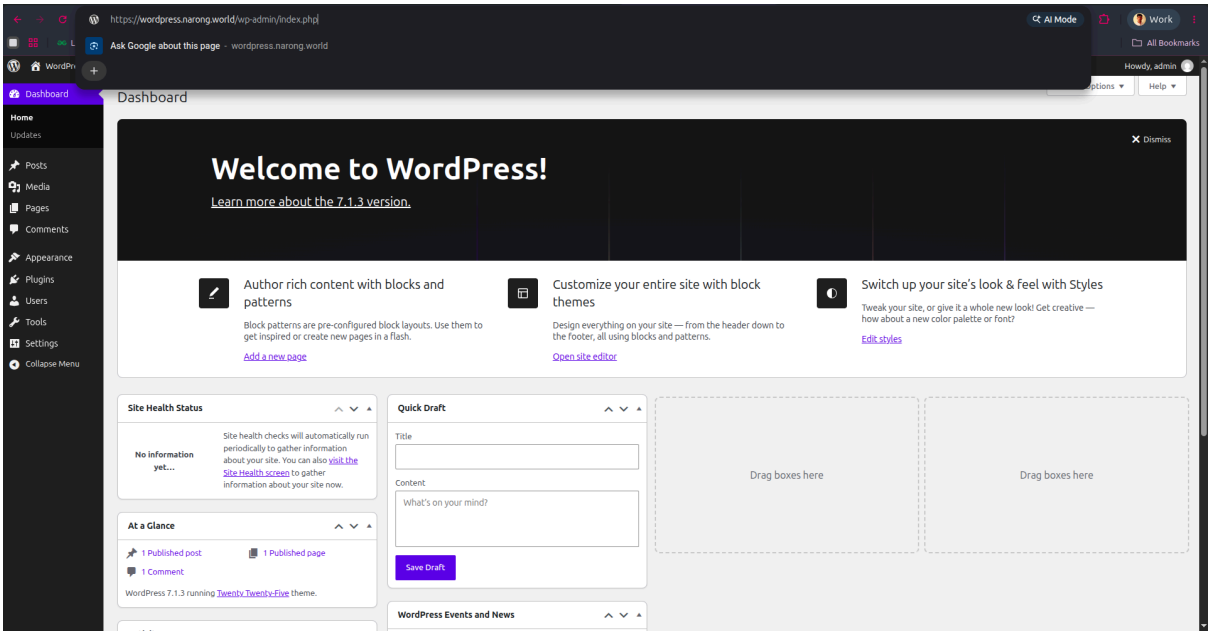
Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/wordpress.narong.world/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/wordpress.narong.world/privkey.pem
This certificate expires on 2027-01-07.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for wordpress.narong.world to /etc/apache2/sites-available/site-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://wordpress.narong.world

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@webserver:/tmp$
```

+ Run Certbot to automatically fetch an SSL certificate and configure Apache to redirect HTTP traffic to HTTPS:

```
sudo certbot --apache -d wordpress.narong.world
```



https://wordpress.narong.world/wp-admin/index.php

Thank you